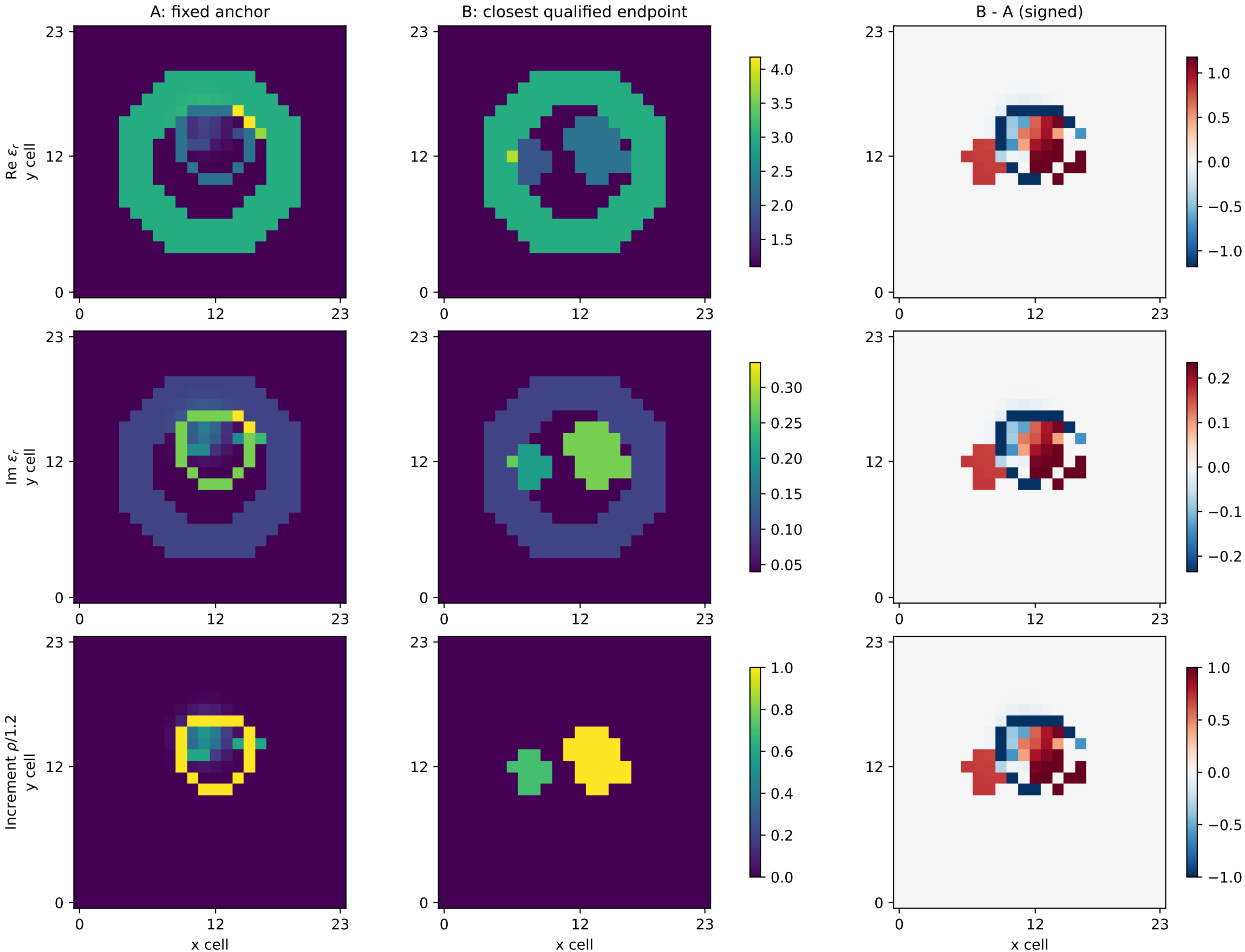
mixed_scale / ring — nearest structurally qualified P0_EXACT pair
d_pred=173.222 (required <=2); same fixed sigma=0.0006465912
32-noise competition: A truth 0/32, B truth 0/32 — NO COMPETING WITNESS



Shape RMS=0.159; occupancy=0.045; boundary=0.042; persistent levels=3/5
Finite bounded search: no global uniqueness claim; no posterior-mass estimate. Shared A/B scales across families.